Khaled Waleed Ba Tais

AI and Computer Vision Engineer | Edge AI

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SUMMARY

Third-year Computer Science (AI) student who ships computer vision on real hardware: Raspberry Pi 5, Hailo-8, NVIDIA Jetson and ESP32. Darb 2026 winner, Top 25 of 225+ teams at GSR Hackathon (KFUPM) and first author of the DragonFruit-MY dataset. Seeking an AI, Computer Vision or Edge AI internship, 19 April to 6 August 2027.

EDUCATION

Taylor's University, Malaysia | BSc (Hons) Computer Science (AI), dual award with UWE Bristol — Sep 2024 to 2027
CGPA 3.73 / 4.00 | Dean's List, 5 consecutive semesters

SKILLS

AI and ML: Python, PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, NumPy, Pandas
Computer Vision: YOLOv5, YOLOv8, OpenCV, MediaPipe, object detection, segmentation, tracking
Edge and IoT: Raspberry Pi 5, Hailo-8, NVIDIA Jetson, ESP32, Arduino, MQTT, ONNX, INT8 quantization
Tools: Git, Linux, Docker, Cloudflare Workers, D1, R2, Wrangler, REST APIs, Hugging Face

PROJECTS

BINA, two-way Arabic Sign Language platform (capstone) — **Prototype** | **Avatars**

- Customer app, admin dashboard, Cloudflare Workers API with D1 and R2, and an ESP32-S3 smartwatch; Saudi and Malaysian avatars from my avatar work with the TALKBIM team.
- 365 automated tests passing; the watch plays signs at 15 fps on real hardware.

ClimaShield, edge AI for farming — **Video** | **Live** | **Code**

- Offline Raspberry Pi decides shade and irrigation from camera and sensors; YOLOv8n leaf screening at 86.4% mAP50.
- Top 25 of 225+ teams at GSR Hackathon (KFUPM) and Top 33 of 439 at the Asian Hackathon for Green Future.

AquaGuard, edge AI drowning early warning — **Live** | **Code**

- YOLO on Raspberry Pi 5 and Hailo-8, no cloud: 90.5% mAP50 person, 94.2% mAP50 pool zone, 8,689 images.
- SAFE, DANGER and EMERGENCY logic with Telegram alerts in under 2 seconds; validated in a real pool test.

SignBridge, Arabic sign language translator — **Demo** | **Code**

- Real-time sign to speech and speech to text; 1st place, Darb 2026, Digital Education track.

RESEARCH

- **DragonFruit-MY**, first author (SSRN preprint, 2026): 727 field-photographed dragon fruits; YOLOv8n mAP@0.5 0.966.
- Paper presenter, 9th Taylor's International SEARCH Conference 2026 (SignBridge).

AWARDS

- **Darb 2026:** 1st place, Digital Education track, MYR 2,000
- **GSR Hackathon 2026, KFUPM:** Top 25 of 225+ teams
- **Asian Hackathon for Green Future 2026:** Top 33 of 439 teams from 22 countries
- **MEWA University Innovation Challenge, Saudi Arabia:** qualified for the next stage
- **MyIPO copyright notification** CRLY2026W05575 for the ClimaShield architecture

CERTIFICATIONS

Claude Academy (Anthropic): Building with the Claude API | Google: Gemini Certified Student | Cisco: IoT and Networking

ACTIVITIES

Student Research Assistant at AI Expo Korea 2026 (Seoul); TV interview at ENERtec Asia 2026; Estidamah interview at GSR 2026; ITEX 2026; LEAP 2026.